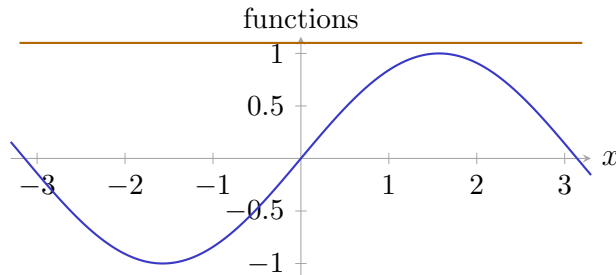


**DYNAMICAL SYSTEMS:
CLASS SEPTEMBER 3, 2026
4. FLOW ON THE CIRCLE.**

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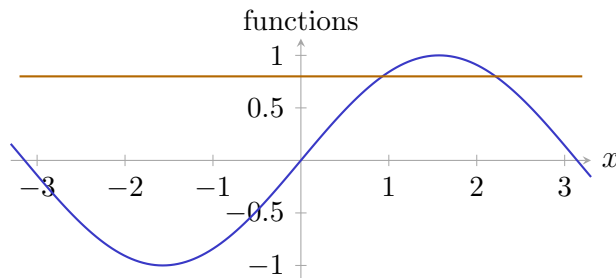
SKILL CHECK PRACTICE

- (1) Assume the time evolution of the phase difference, ϕ , between an oscillator and a reference signal is given by the system $\dot{\phi} = 1.1 - \sin \phi$.



What is the long term behavior of the phase difference in this system? *If it approaches a fixed value, provide an estimate of that value.*

Extra example: $\dot{\phi} = 0.8 - \sin \phi$



- (2) Draw the phase portrait for a 1D differential equation both on the real line and also on the circle.

SKILL CHECK PRACTICE SOLUTION

Answer: the phase difference is drifting (always increasing).

Explanation (not needed on the skill check itself):

This question requires us to interpret ϕ as a phase difference, rather than as the phase of a single oscillator.

If there were an intersection between the sinusoid and the straight line then the phase difference would approach a fixed value (that you could identify) associated with a stable or half-stable fixed point.

In this picture, though, $\dot{\phi} = 1.1 - \sin \phi$ and $\sin \phi < 1.1$ for all values of ϕ . So $\dot{\phi} > 0$ for all phase differences, ϕ . This means that the **phase difference is always changing**. In some sense, it is always increasing ($\dot{\phi} > 0$, after all). However, when the phase difference passes through $2\pi n$ for n an integer, the oscillator and the reference momentarily have the same phase angle, so if we look at the two oscillators on a circle, one of them will appear to ‘lap’ the other one over and over again.

PERIOD OF THE OSCILLATION: SEE SECTION 4.3

Consider the system $\dot{\theta} = \omega - a \sin \theta$. Integrate the time over one oscillation period T :

$$T = \int_0^T dt = \int_0^{2\pi} \frac{dt}{d\theta} d\theta = \int_0^{2\pi} \frac{1}{\omega - a \sin \theta} d\theta$$

Here we first made a change of variables in the integration from t to θ , and then noted that $dt/d\theta = 1/\dot{\theta}$.

TEAMS

Post screenshots of your work to the course Google Drive today: Link to drive folder. Include words, labels, and other short notes on your whiteboard that might make those solutions useful to you or your classmates. Name your images using the format:

- **teamXX-YY.EXT**

where XX is your two-digit team number (e.g. 06 if you are team 6), YY is the two digit number of image for today (e.g. 01 for the first image, 02 for the second, and so on.), and EXT is the extension for the image filetype you used.

TEAM ACTIVITY

- (1) (4.1.1) For which values of a does the equation $\dot{\theta} = \sin(a\theta)$ give a well-defined vector field on the circle? Recall that, in order to be well defined on the circle, the flow must repeat itself as you make a full revolution around the circle.

For $a = 3$, find and classify all the fixed points and sketch the phase portrait on the circle. It may help to make a plot of $\sin(3\theta)$.

Solution. For the vector field to be well-defined on the circle, it must be 2π -periodic:

$$\sin(a(\theta + 2\pi)) = \sin(a\theta) \quad \text{for every } \theta.$$

Using the angle-addition formula,

$$\sin(a\theta + 2\pi a) = \sin(a\theta) \cos(2\pi a) + \cos(a\theta) \sin(2\pi a).$$

This equals $\sin(a\theta)$ for every θ precisely when

$$\cos(2\pi a) = 1 \quad \text{and} \quad \sin(2\pi a) = 0.$$

Therefore,

$$\boxed{a \in \mathbb{Z}}.$$

For $a = 3$, the fixed points satisfy

$$\sin(3\theta) = 0,$$

and hence, modulo 2π ,

$$\theta = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{5\pi}{3}.$$

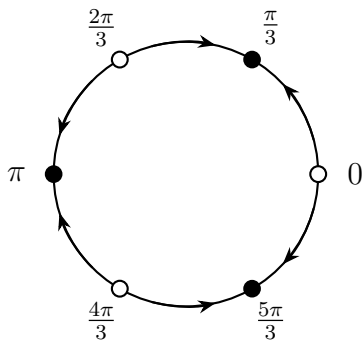
Let $f(\theta) = \sin(3\theta)$. Since

$$f'(\theta) = 3 \cos(3\theta), \quad f'\left(\frac{k\pi}{3}\right) = 3(-1)^k,$$

the fixed points with even k are unstable, while those with odd k are stable. Thus

unstable: $0, \frac{2\pi}{3}, \frac{4\pi}{3},$ stable: $\frac{\pi}{3}, \pi, \frac{5\pi}{3}.$

The arrows point away from each open (unstable) point and toward the neighboring filled (stable) points:



□

(2) (4.3.3) For $\dot{\phi} = \mu \sin \phi - \sin 2\phi$:

- (a) Check that the vector field is well-defined on the circle
- (b) Draw phase portraits for:
 - large positive μ
 - $\mu = 0$
 - large negative μ
- (c) Use $\sin 2\phi = 2 \cos \phi \sin \phi$ to find mathematical expressions for fixed points.
- (d) Classify the bifurcations that occur as μ varies.
- (e) Find the bifurcation values of μ .
- (f) Think of ϕ as describing the **phase of a single oscillator**. For what values of μ is the system “oscillating”?
- (g) Think of ϕ as describing the **phase difference** between an oscillator and a reference. For what values of μ is the oscillator entrained (phase-locked) to the reference? What sets their phase difference?

Solution. Write

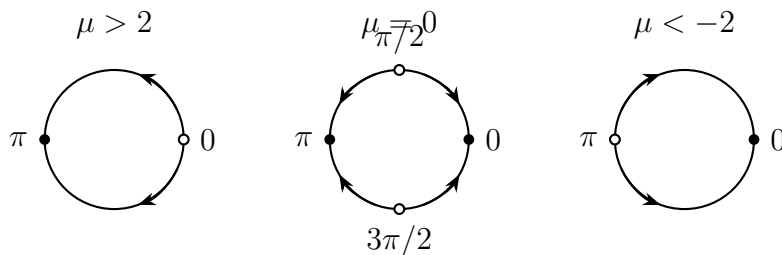
$$F(\phi; \mu) = \mu \sin \phi - \sin 2\phi = \sin \phi (\mu - 2 \cos \phi).$$

- (a) Since both sine terms are 2π -periodic,

$$F(\phi + 2\pi; \mu) = \mu \sin(\phi + 2\pi) - \sin(2\phi + 4\pi) = F(\phi; \mu).$$

Thus the vector field is well-defined on the circle for every $\mu \in \mathbb{R}$.

- (b) The phase portraits for the three requested parameter regimes are shown below. Open circles are unstable fixed points and filled circles are stable fixed points.



- (c) The fixed-point equation is

$$\sin \phi (\mu - 2 \cos \phi) = 0.$$

Hence $\phi = 0$ and $\phi = \pi$ are fixed points for every μ . When $|\mu| \leq 2$, there are also fixed points satisfying

$$\cos \phi = \frac{\mu}{2},$$

namely

$$\phi = \arccos\left(\frac{\mu}{2}\right) \quad \text{and} \quad \phi = 2\pi - \arccos\left(\frac{\mu}{2}\right) \pmod{2\pi}.$$

To determine stability, differentiate:

$$F_\phi(\phi; \mu) = \mu \cos \phi - 2 \cos 2\phi.$$

Therefore

$$F_\phi(0; \mu) = \mu - 2, \quad F_\phi(\pi; \mu) = -\mu - 2.$$

Thus 0 is stable for $\mu < 2$ and unstable for $\mu > 2$, while π is unstable for $\mu < -2$ and stable for $\mu > -2$. At either additional fixed point, where $\cos \phi = \mu/2$,

$$F_\phi = 2 - \frac{\mu^2}{2} > 0 \quad (|\mu| < 2),$$

so both additional fixed points are unstable.

- (d) As μ passes through 2, the two unstable off-axis fixed points collide with $\phi = 0$, while the stability of $\phi = 0$ changes. This is a subcritical pitchfork bifurcation. Likewise, at $\mu = -2$ the two unstable fixed points collide with $\phi = \pi$, producing a second subcritical pitchfork bifurcation.
- (e) The bifurcation values are therefore

$$\boxed{\mu = -2 \quad \text{and} \quad \mu = 2}.$$

- (f) If ϕ is the phase of a single oscillator, oscillation would require the phase to keep moving around the circle. This never occurs, because the system has fixed points for every μ . Hence there are

$$\boxed{\text{no values of } \mu \text{ for which the system oscillates.}}$$

- (g) If ϕ is a phase difference, the oscillator is entrained whenever the phase difference approaches or remains at a fixed point. There is a stable fixed point for every μ , so the oscillator is entrained for every initial condition: generic initial conditions approach a stable equilibrium, while an initial condition exactly at an unstable equilibrium remains phase-locked there but is not robust to perturbations. More specifically,

parameter range	asymptotically stable phase difference(s)
$\mu < -2$	0
$-2 < \mu < 2$	0 or π
$\mu > 2$	π

with the same limiting choices at $\mu = -2$ and $\mu = 2$ determined by the remaining stable equilibrium. In the bistable range $-2 < \mu < 2$, the initial phase difference determines whether the system approaches 0 or π ; the two unstable fixed points form the basin boundaries.

□

Answers.

1: $\sin(a(\theta + 2\pi)) = \sin(a\theta + a2\pi) = \sin(a\theta)\cos(2\pi a) + \cos(a\theta)\sin(2\pi a)$. When a is an integer, this will lead to $\sin(a(\theta + 2\pi)) = \sin(a\theta)$. Otherwise, there's a non-zero cosine contribution, which would be a problem.

2a: $\mu \sin(\phi + 2\pi) - \sin(2\phi + 4\pi) = \mu \sin \phi - \sin 2\phi$, well-defined. 2b: 2c: $\dot{\phi} = \mu \sin \phi - 2 \sin \phi \cos \phi = \sin \phi(\mu - 2 \cos \phi)$ so $\phi = 0, \pi$ from $\sin \phi = 0$ and $\cos \phi = \mu/2$, which will have zeros for $-2 \leq \mu \leq 2$. 2d: Two subcritical pitchfork bifurcations. 2e: $\mu \sin \phi$ is tangent to $\sin 2\phi$ at the 0 fixed point when $\mu\phi$ (the linear approximation) is equal to 2ϕ , so when $\mu = 2$. At π the tangency occurs when $\mu = -2$. So a subcritical pitchfork for $\phi = 0$ and $\mu = 2$ and a subcritical pitchfork for $\phi = \pi$ at $\mu = -2$. 2f: no oscillation ever. 2g: there is always entrainment (there is always a fixed point).